

## SAN SEBASTIAN IGUELDO, Spain

WMO: 080270

Lat: 43.3064N

Lon: 2.0411W

Elev: 850

StdP: 14.25

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 31.8       | 34.4 | 23.1                          | 18.0 | 39.7 | 26.2 | 20.9 | 40.6 | 38.8                  | 52.4 | 34.4 | 51.8 | 5.9                   | 160  | 0.489 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.4                    | 83.1            | 68.7 | 78.8 | 66.8 | 75.2 | 65.7 | 71.4                | 78.5 | 69.5 | 74.7 | 68.1 | 72.3 | 11.1                 | 160  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.4                            | 111.6 | 74.0 | 67.9 | 105.8 | 71.8 | 66.6 | 101.2 | 70.2 | 35.7          | 78.6 | 34.1 | 74.8 | 32.9 | 72.4 | 77.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 31.7          | 26.9 | 23.0 | DB                         | 28.8 | 93.4               | 3.1 | 4.3                                                | 26.5 | 96.4       | 24.7 | 98.9       | 23.0 | 101.3      | 20.7 | 104.4 | (4) |
| (5)               |      |      | WB                         | 26.7 | 75.9               | 3.5 | 1.5                                                | 24.2 | 77.0       | 22.2 | 77.8       | 20.2 | 78.7       | 17.7 | 79.8  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 56.6        | 47.3  | 47.6  | 50.6  | 53.2  | 57.8  | 63.0  | 66.0  | 67.1  | 63.9  | 60.4  | 52.6  | 48.8  | (6)  |  |
|                                                                        | DBStd       | 9.12        | 6.21  | 6.82  | 6.45  | 5.61  | 5.56  | 5.34  | 4.20  | 4.22  | 4.87  | 6.38  | 6.30  | 6.37  | (7)  |  |
|                                                                        | HDD50       | 476         | 125   | 112   | 69    | 23    | 3     | 0     | 0     | 0     | 0     | 4     | 42    | 98    | (8)  |  |
|                                                                        | HDD65       | 3353        | 550   | 487   | 448   | 357   | 236   | 99    | 32    | 20    | 77    | 171   | 373   | 503   | (9)  |  |
|                                                                        | CDD50       | 2878        | 40    | 46    | 87    | 120   | 244   | 390   | 497   | 529   | 418   | 327   | 120   | 60    | (10) |  |
|                                                                        | CDD65       | 281         | 0     | 0     | 1     | 4     | 12    | 39    | 65    | 85    | 45    | 29    | 1     | 0     | (11) |  |
|                                                                        | CDH74       | 1089        | 0     | 0     | 0     | 21    | 79    | 260   | 213   | 305   | 164   | 46    | 1     | 0     | (12) |  |
|                                                                        | CDH80       | 297         | 0     | 0     | 0     | 2     | 18    | 90    | 73    | 88    | 25    | 1     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 9.3         | 11.1  | 10.9  | 10.5  | 9.7   | 8.8   | 7.4   | 6.9   | 6.6   | 7.5   | 9.6   | 11.2  | 11.0  | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 62.9        | 6.2   | 5.0   | 4.6   | 5.1   | 4.8   | 4.0   | 3.6   | 4.6   | 4.1   | 5.4   | 8.3   | 5.7   | (15) |  |
|                                                                        | PrecMax     | 84.5        | 14.6  | 12.7  | 9.4   | 10.5  | 10.4  | 10.1  | 7.4   | 11.0  | 8.7   | 12.9  | 15.9  | 10.0  | (16) |  |
|                                                                        | PrecMin     | 49.2        | 0.2   | 0.7   | 1.3   | 1.9   | 2.4   | 0.9   | 1.5   | 2.2   | 1.5   | 2.0   | 1.8   | 0.6   | (17) |  |
|                                                                        | PrecStd     | 9.0         | 3.7   | 3.3   | 2.2   | 2.4   | 2.4   | 2.1   | 1.8   | 2.6   | 1.7   | 3.0   | 3.4   | 2.8   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 64.0  | 67.3  | 71.2  | 77.0  | 82.2  | 89.3  | 88.7  | 87.3  | 82.8  | 78.2  | 69.3  | 64.9  | (19) |  |
|                                                                        |             | MCWB        | 52.0  | 51.0  | 54.6  | 60.8  | 64.3  | 70.6  | 70.5  | 72.0  | 68.4  | 64.9  | 57.9  | 53.9  | (20) |  |
|                                                                        | 2%          | DB          | 60.3  | 62.9  | 67.6  | 70.3  | 74.9  | 81.1  | 79.4  | 81.8  | 78.5  | 74.7  | 66.5  | 62.0  | (21) |  |
|                                                                        |             | MCWB        | 52.0  | 50.2  | 53.6  | 57.3  | 61.5  | 66.8  | 69.4  | 69.7  | 65.9  | 63.1  | 56.7  | 52.8  | (22) |  |
|                                                                        | 5%          | DB          | 58.0  | 59.5  | 64.1  | 65.4  | 70.4  | 74.9  | 74.3  | 76.5  | 74.6  | 72.3  | 64.1  | 59.4  | (23) |  |
|                                                                        |             | MCWB        | 50.5  | 49.5  | 52.0  | 54.6  | 59.3  | 65.1  | 68.4  | 68.3  | 64.9  | 61.5  | 55.8  | 51.4  | (24) |  |
|                                                                        | 10%         | DB          | 55.5  | 56.4  | 60.5  | 61.6  | 66.5  | 69.9  | 71.2  | 72.7  | 70.9  | 69.7  | 61.4  | 57.2  | (25) |  |
|                                                                        |             | MCWB        | 49.3  | 48.4  | 50.5  | 53.2  | 58.3  | 64.0  | 67.1  | 67.7  | 64.5  | 60.4  | 54.1  | 50.3  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 55.2  | 54.8  | 57.2  | 61.8  | 66.6  | 72.7  | 73.6  | 74.3  | 71.1  | 66.8  | 60.9  | 56.5  | (27) |  |
|                                                                        |             | MCDB        | 60.4  | 61.5  | 67.5  | 74.9  | 78.7  | 85.6  | 81.1  | 81.5  | 78.1  | 73.6  | 64.8  | 60.6  | (28) |  |
|                                                                        | 2%          | WB          | 53.4  | 53.0  | 54.9  | 58.3  | 63.3  | 68.3  | 70.8  | 71.7  | 68.8  | 64.6  | 58.5  | 54.4  | (29) |  |
|                                                                        |             | MCDB        | 57.5  | 58.3  | 64.2  | 66.5  | 70.6  | 75.8  | 76.9  | 78.0  | 74.1  | 70.9  | 63.6  | 58.8  | (30) |  |
|                                                                        | 5%          | WB          | 51.9  | 51.4  | 53.2  | 56.3  | 61.2  | 66.4  | 69.0  | 69.8  | 67.2  | 63.2  | 56.9  | 52.9  | (31) |  |
|                                                                        |             | MCDB        | 55.6  | 56.2  | 60.4  | 62.3  | 67.2  | 72.4  | 73.2  | 74.3  | 71.6  | 69.0  | 61.8  | 57.5  | (32) |  |
|                                                                        | 10%         | WB          | 50.4  | 50.0  | 51.8  | 54.7  | 59.6  | 64.7  | 67.5  | 68.3  | 65.7  | 62.0  | 55.5  | 51.6  | (33) |  |
|                                                                        |             | MCDB        | 54.0  | 54.2  | 57.5  | 59.2  | 64.4  | 69.1  | 70.7  | 71.8  | 69.5  | 67.3  | 59.9  | 56.2  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 7.0   | 8.3   | 8.8   | 8.9   | 9.1   | 8.7   | 7.7   | 8.4   | 8.7   | 8.6   | 7.2   | 7.2   | (35) |  |
|                                                                        |             | MCDBR       | 9.6   | 12.6  | 15.0  | 16.5  | 18.3  | 19.3  | 15.1  | 16.7  | 15.3  | 12.5  | 9.9   | 9.7   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 5.9   | 6.8   | 7.7   | 8.3   | 9.1   | 9.5   | 8.0   | 8.7   | 8.1   | 6.6   | 5.6   | 6.1   | (37) |  |
|                                                                        |             | MCWBR       | 8.2   | 10.4  | 11.8  | 13.4  | 14.6  | 15.1  | 12.3  | 12.8  | 12.0  | 10.8  | 8.6   | 8.4   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.320 | 0.335 | 0.367 | 0.390 | 0.400 | 0.411 | 0.402 | 0.401 | 0.380 | 0.358 | 0.335 | 0.316 | (40) |  |
|                                                                        |             | taud        | 2.514 | 2.462 | 2.380 | 2.325 | 2.334 | 2.335 | 2.363 | 2.375 | 2.424 | 2.488 | 2.503 | 2.521 | (41) |  |
|                                                                        | Ebn at Noon |             | 259   | 273   | 276   | 278   | 277   | 273   | 274   | 271   | 269   | 261   | 250   | 249   | (42) |  |
|                                                                        |             | Edh at Noon | 23    | 28    | 34    | 38    | 39    | 39    | 38    | 36    | 32    | 27    | 23    | 20    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 450   | 717   | 1106  | 1442  | 1718  | 1903  | 1911  | 1688  | 1357  | 881   | 526   | 420   | (44) |  |
|                                                                        | RadStd      |             | 55    | 95    | 138   | 150   | 134   | 121   | 100   | 103   | 70    | 73    | 64    | 51    | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) Regional (14 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page